



Name: Cohort/Batch: Date: Score:

IBM DB2 PRACTICE SHEET - 10 CONCEPTUAL Q&A

10 conceptual Q&A Databases

TOTAL: 10 QUESTIONS

Q1 What type of database is IBM Db2 and what are its core strengths? Core Model

Q6 How do you monitor and tune slow queries in IBM Db2? Performance

Q2 How does IBM Db2 guarantee consistency and handle transactions? Architecture

Q7 What backup and restore strategies does IBM Db2 support? Backup

Q3 What index types does IBM Db2 support and when do you use each? Indexing

Q8 How do you handle schema migrations in IBM Db2? Compliance

Q4 How do you design a schema or data model in IBM Db2? Modeling

Q9 What security features does IBM Db2 provide? Security

Q5 How does IBM Db2 scale horizontally? SSO / IAP

Q10 What are common anti-patterns when using IBM Db2 in production? Tuning



ANSWER KEY & EXPLANATORY SOLUTIONS

Comprehensive verified answers, best-practice configs, security controls & reference

VERIFIED SOLUTIONS

A1 IBM Db2 is a relational, document, key-value,

Mitigation

IBM Db2 is a relational, document, key-value, graph, or time-series store. Its strengths such as ACID guarantees, horizontal scale, or flexible schema guide the workloads it is best suited for.

A6 IBM Db2 provides a slow query log,

Observability

IBM Db2 provides a slow query log, explain/explain plan, or profiling tools to surface inefficient queries. Common fixes include adding indexes, rewriting queries, or increasing cache size.

A2 IBM Db2 provides either full ACID transactions

Primitives

IBM Db2 provides either full ACID transactions or eventual consistency depending on its CAP theorem positioning. Transaction support varies from none (simple K/V stores) to serializable isolation (relational DBs).

A7 IBM Db2 supports logical backups (export SQL

Automation

IBM Db2 supports logical backups (export SQL or BSON), physical backups (filesystem snapshot), and continuous replication to a standby. Point-in-time recovery requires WAL/oplog archiving on top of backups.

A3 IBM Db2 offers B-tree, hash, full-text, spatial,

Hardening

IBM Db2 offers B-tree, hash, full-text, spatial, or composite indexes depending on the engine. Choose based on query patterns: B-tree for range queries, hash for equality lookups, full-text for search.

A8 Schema migrations in IBM Db2 are managed

Governance

Schema migrations in IBM Db2 are managed with versioned migration files applied in order by a migration tool. Migrations should be idempotent and tested in staging before production to avoid downtime.

A4 Schema design in IBM Db2 involves normalizing

Gotchas

Schema design in IBM Db2 involves normalizing relations (relational DB) or denormalizing for access patterns (document/key-value). Understanding query needs upfront prevents costly migrations later.

A9 IBM Db2 supports role-based access control, encrypted

Assessment

IBM Db2 supports role-based access control, encrypted connections (TLS), encryption at rest, and audit logging. Always restrict user privileges to the minimum needed and disable anonymous access in production.

A5 IBM Db2 scales via replication (read replicas)

Federation

IBM Db2 scales via replication (read replicas) for read-heavy workloads and sharding (partitioning data by key) for write-heavy ones. Some IBM Db2 implementations handle this automatically; others require manual configuration.

A10 Common mistakes include selecting all columns instead,

Optimization

Common mistakes include selecting all columns instead of needed fields, missing indexes on frequently queried columns, running migrations without backups, and storing large blobs directly in the database instead of object storage.